

SERVER SIDE MODULE

CONTENTS

This module has the following files:

1. SERVER MODULE PHASE 1.docx
2. SERVER MODULE PHASE 1.pdf
3. SERVER MODULE PHASE 1 MEDIA.zip

INTRODUCTION

The client has approached you to build a backend system and frontend web application for an Inventory Management Application called **StokPintar**. The system must allow users to register, manage their products, record stock-in and stock-out movements, and view basic stock reports.

PROJECT OVERVIEW

This project consists of two main phases: In Phase 1, you are required to develop a REST API using the Laravel Framework, implementing core features such as user authentication, product management, stock movement recording and stock reporting. In Phase 2, you will build a frontend web application that integrates with the API, using either React or Vue as your framework of choice.

To support your work, media files have been provided, including Postman Collection and Environment, a Database Dump (.sql), and Framework Scaffolding.

PHASE 1 - Backend Development (REST API)

A. Authentication

There are three endpoints: register, login, and logout. Sanctum should be used to create tokens. The generated token will be used to access all available endpoints. The logout function should only deactivate the token from the specific device making the request.

A1: Register

POST /api/auth/register

For users to create an account in the application.

Request Headers

Key	Value
Accept	application/json

Body (JSON)

Field	Rules
full_name	required.
email	required, valid email format, unique.
password	required, min 6 chars.

A1a: Response Success

Key	Value
Status Code	201
	{
	"status": "success",
	"message": "Registration successful",
	"data": {
	"name": "Dedi",
	"email": "dedi@webtech.id",
	"updated_at": "2025-07-11T16:01:46.000000Z",
	"created_at": "2025-07-11T16:01:46.000000Z",
	"id": 6,
	"token": "1 wwwOrgoVe4Xyj5dvUmnf6WTPTaA..."
	}
	}

A1b: Response Invalid Field

Key	Value
Status Code	422
	{
	"status": "error",
	"message": "Invalid field",
	"errors": {
	"name": [
	"The name field is required."
	],
	"email": [
	"The email has already been taken."
	],
	"password": [
	"The password field must be at least 6 characters."
	]
	}
	}

A2: Login

POST /api/auth/login

For users login into the application using email and password.

Request Headers

Key	Value
Accept	application/json

Body (JSON)

Field	Rules
email	required.
password	required.

A2a: Response Success

Key	Value
Status Code	200
	{ "status": "success", "message": "Login successful", "data": { "id": 1, "name": "Budi", "email": "budi@webtech.id", "created_at": "2025-07-11T16:01:38.000000Z", "updated_at": "2025-07-11T16:01:38.000000Z", "token": "2 0Q2TAdua0FOslp1F4cxfrzQgkp..." } }

A2b: Response Failed

Key	Value
Status Code	401
	{ "status": "error", "message": "Username or password incorrect" }

A3: Logout

POST /api/auth/logout

For logged-in user logout from the application. It specifically deactivates the authentication token associated with the current device making the request.

Request Headers

Key	Value
Accept	application/json
Authorization	Bearer <TOKEN>

A3a: Response Success

Key	Value
Status Code	200
{	
"status": "success",	
"message": "Logout successful"	
}	

A3b: Response Invalid Token

Key	Value
Status Code	401
{	
"status": "error",	
"message": "Unauthenticated."	
}	

B. Unit and Category

The application uses a Category system to classify stock movements. Each category must be assigned a type, which can either be:

- **IN** – for movements that increase the product stock (e.g., Purchase, Customer Return, Initial Stock)
- **OUT** – for movements that decrease the product stock (e.g., Sale, Damaged, Internal Use)

Categories help users organize their stock activities and are required when creating a stock movement.

B1: Get All Units

GET /api/units

For logged-in users to get all available units.

Request Headers

Key	Value
Accept	application/json
Authorization	Bearer <TOKEN>

B1a: Response Success

Key	Value
-----	-------

Status Code	200
-------------	-----

```
{
  "status": "success",
  "message": "Get all units successful",
  "data": {
    "units": [
      {
        "id": 1,
        "name": "Pieces",
        "symbol": "pcs",
        "code": "PCS",
        "created_at": "2025-07-11T16:01:37.000000Z",
        "updated_at": "2025-07-11T16:01:37.000000Z"
      },
      "..."
    ]
  }
}
```

B1b: Response Invalid Token

Key	Value
-----	-------

Status Code	401
-------------	-----

```
{
  "status": "error",
  "message": "Unauthenticated."
}
```

B2: Get All Categories

GET /api/categories

For logged-in users to get all available categories.

Request Headers

Key	Value
-----	-------

Accept	application/json
--------	------------------

Authorization	Bearer <TOKEN>
---------------	----------------

B2a: Response Success

Key	Value
-----	-------

Status Code	200
-------------	-----

```
{
  "status": "success",
  "message": "Get all categories successful",
}
```

```

"data": {
  "categories": [
    {
      "id": 1,
      "name": "Purchase",
      "icon": "📦",
      "color": "#00B894",
      "type": "IN",
      "created_at": "2025-07-11T16:01:38.000000Z",
      "updated_at": "2025-07-11T16:01:38.000000Z"
    },
    "..."
  ]
}

```

B2b: Response Invalid Token

Key	Value
Status Code	401

```

{
  "status": "error",
  "message": "Unauthenticated."
}

```

C. Product

A Product is a stock container used to track the quantity of an item. Users can create multiple products to organize their inventory (e.g., raw materials, finished goods, or merchandise).

Each product must display a `current_stock`, which is automatically and accurately calculated based on all associated stock movements.

- **IN** movements will increase the stock.
- **OUT** movements will decrease the stock.

The system must ensure that the `current_stock` is always up-to-date and reflects the correct total from all movements made on that product.

C1: Add Product

POST `/api/products`

For logged-in users to add a product.

Request Headers

Key	Value
-----	-------

Key	Value
Accept	application/json
Authorization	Bearer <TOKEN>

Body (JSON)

Field	Rules
name	required.
unit_code	required, valid unit_code data.

C1a: Response Success

Key	Value
Status Code	201
	{
	"status": "success",
	"message": "Product added successful",
	"data": {
	"name": "Beras Premium 5kg",
	"user_id": 1,
	"updated_at": "2025-07-11T16:22:44.000000Z",
	"created_at": "2025-07-11T16:22:44.000000Z",
	"id": 26,
	"unit_code": "PCK"
	}
	}

C1b: Response Invalid Field

Key	Value
Status Code	422
	{
	"status": "error",
	"message": "Invalid field",
	"errors": {
	"name": [
	"The name field is required."
	],
	"unit_code": [
	"The selected unit code is invalid."
	]
	}
	}

C1c: Response Invalid Token

Key	Value
Status Code	401

```
{
  "status": "error",
  "message": "Unauthenticated."
}
```

C2: Update Product

PUT /api/products/:productId

For logged-in users to update their own product.

Request Headers

Key	Value
Accept	application/json
Authorization	Bearer <TOKEN>

Body (JSON)

Field	Rules
name	required.

C2a: Response Success

Key	Value
Status Code	200
{	
"status": "success",	
"message": "Product updated successful",	
"data": {	
"id": 1,	
"user_id": 1,	
"name": "Indomie Goreng",	
"created_at": "2025-07-11T16:01:38.000000Z",	
"updated_at": "2025-07-11T16:26:25.000000Z",	
"deleted_at": null ,	
"unit_code": "PCS"	
}	
}	

C2b: Response Not Found

Key	Value
Status Code	404
{	
"status": "error",	
"message": "Not found"	
}	

C2c: Response Forbidden

Key	Value
Status Code	403
{	
"status": "error",	
"message": "Forbidden access"	
}	

C2d: Response Invalid Token

Key	Value
Status Code	401
{	
"status": "error",	
"message": "Unauthenticated."	
}	

C3: Delete Product

DELETE /api/products/:productId

For logged-in users to delete their own product.

Request Headers

Key	Value
Accept	application/json
Authorization	Bearer <TOKEN>

C3a: Response Success

Key	Value
Status Code	200
{	
"status": "success",	
"message": "Product deleted successful"	
}	

C3b: Response Not Found

Key	Value
Status Code	404
{	
"status": "error",	
"message": "Not found"	
}	

C3c: Response Forbidden

Key	Value
-----	-------

Key	Value
Status Code	403
{	
"status": "error",	
"message": "Forbidden access"	
}	

C3d: Response Invalid Token

Key	Value
Status Code	401
{	
"status": "error",	
"message": "Unauthenticated."	
}	

C4: Get All Products

GET /api/products

For logged-in users to get all of their own products.

Request Headers

Key	Value
Accept	application/json
Authorization	Bearer <TOKEN>

C4a: Response Success

Key	Value
Status Code	200
{	
"status": "success",	
"message": "Get all products successful",	
"data": {	
"products": [	
{	
"id": 2,	
"user_id": 1,	
"name": "Indomie Goreng",	
"created_at": "2025-07-11T16:01:38.000000Z",	
"updated_at": "2025-07-11T16:01:38.000000Z",	
"deleted_at": null,	
"unit_code": "PCS",	
"current_stock": 1225	
},	
"..."	
]	
}	

```
}  
}
```

C4b: Response Invalid Token

Key	Value
-----	-------

Status Code	401
-------------	-----

```
{  
  "status": "error",  
  "message": "Unauthenticated."  
}
```

C5: Get Detail Product

GET /api/products/:productId

For logged-in users to get details about their own products.

Request Headers

Key	Value
-----	-------

Accept	application/json
--------	------------------

Authorization	Bearer <TOKEN>
---------------	----------------

C5a: Response Success

Key	Value
-----	-------

Status Code	200
-------------	-----

```
{  
  "status": "success",  
  "message": "Get detail product successful",  
  "data": {  
    "id": 1,  
    "user_id": 1,  
    "name": "Beras Premium 5kg",  
    "created_at": "2025-07-11T16:01:38.000000Z",  
    "updated_at": "2025-07-11T16:26:25.000000Z",  
    "deleted_at": null,  
    "unit_code": "PCK",  
    "current_stock": 486  
  }  
}
```

C5b: Response Not Found

Key	Value
-----	-------

Status Code	404
-------------	-----

```
{  
  "status": "error",  
  "message": "Not found"  
}
```

C5c: Response Forbidden

Key	Value
Status Code	403
{	
"status": "error",	
"message": "Forbidden access"	
}	

C5d: Response Invalid Token

Key	Value
Status Code	401
{	
"status": "error",	
"message": "Unauthenticated."	
}	

D. Stock Movement

A Stock Movement represents an inventory record that occurs on a specific product. Each movement must be linked to a product and assigned to a category, which determines whether it is a stock IN or OUT.

Key attributes of a stock movement include:

- **Product** – the product where the movement belongs
- **Category** – defines the type (IN or OUT) based on the selected category
- **Quantity** – the amount of stock moved (positive integer)
- **Date** – the date when the movement occurred
- **Note** (*optional*) – a short note about the movement

The movement's impact on the product's `current_stock` depends on the type defined by the selected category:

- **IN** category → increases the `current_stock`
- **OUT** category → decreases the `current_stock`

The system must ensure that movements are accurately recorded and that the `current_stock` of each product is updated accordingly.

D1: Add Stock Movement

POST /api/stock-movements

For logged-in users to add a stock movement.

Request Headers

Key	Value
Accept	application/json
Authorization	Bearer <TOKEN>

Body (JSON)

Field	Rules
product_id	required, valid product_id data.
category_id	required, valid category_id data.
quantity	required, integer, greater or equal 1
date	required, date format "Y-m-d"
note	optional

D1a: Response Success

Key	Value
Status Code	201
	<pre>{ "status": "success", "message": "Stock movement added successful", "data": { "category_id": 3, "product_id": 1, "quantity": 50, "note": "Pengiriman dari supplier ABC", "date": "2025-07-31", "updated_at": "2025-07-11T16:44:09.000000Z", "created_at": "2025-07-11T16:44:09.000000Z", "id": 842 } }</pre>

D1b: Response Invalid Field

Key	Value
Status Code	422
	<pre>{ "status": "error", "message": "Invalid field", "errors": { "product_id": ["The product id field is required."], "category_id": ["The selected category id is invalid."], "quantity": ["The quantity field must be at least 1."] } }</pre>

```
]
}
```

D1c: Response Invalid Token

Key	Value
Status Code	401
{	
"status": "error",	
"message": "Unauthenticated."	
}	

D2: Delete Stock Movement

DELETE /api/stock-movements/:movementId

For logged-in users to delete their own stock movement.

Request Headers

Key	Value
Accept	application/json
Authorization	Bearer <TOKEN>

D2a: Response Success

Key	Value
Status Code	200
{	
"status": "success",	
"message": "Stock movement deleted successful"	
}	

D2b: Response Not Found

Key	Value
Status Code	404
{	
"status": "error",	
"message": "Not found"	
}	

D2c: Response Forbidden

Key	Value
Status Code	403
{	
"status": "error",	
"message": "Forbidden access"	
}	

D2d: Response Invalid Token

Key	Value
Status Code	401
{	
"status": "error",	
"message": "Unauthenticated."	
}	

D3: Get Stock Movements

GET /api/stock-movements

For logged-in users to get all of their own stock movements ordered by movement date descending.

Request Headers

Key	Value
Accept	application/json
Authorization	Bearer <TOKEN>

Request Parameters

Key	Rules
page	optional, integer, default 1
per_page	optional, integer, default 25
month	optional, integer, between 1 and 12
year	optional, integer

D3a: Response Success

Key	Value
Status Code	200
{	
"current_page": 1,	
"data": [	
{	
"id": 842,	
"category_id": 3,	
"product_id": 1,	
"quantity": 50,	
"note": "Pengiriman dari supplier ABC",	
"date": "2025-07-31",	
"created_at": "2025-07-11T16:44:09.000000Z",	
"updated_at": "2025-07-11T16:44:09.000000Z",	
"product": {	
"id": 1,	
"user_id": 1,	

```

        "name": "Beras Premium 5kg",
        "created_at": "2025-07-11T16:01:38.000000Z",
        "updated_at": "2025-07-11T16:26:25.000000Z",
        "deleted_at": null,
        "unit_code": "PCK"
    },
    "category": {
        "id": 3,
        "name": "Sale",
        "icon": "🛒",
        "type": "OUT",
        "created_at": "2025-07-11T16:01:38.000000Z",
        "updated_at": "2025-07-11T16:01:38.000000Z"
    }
},
"..."
],
"from": 1,
"last_page": 170,
"per_page": 1,
"to": 1,
"total": 170
}

```

D3b: Response Invalid Token

Key	Value
Status Code	401

```

{
  "status": "error",
  "message": "Unauthenticated."
}

```

E. Stock Report

E1: Get Summary by OUT Category

GET /api/reports/summary-by-category/out

For logged-in users to get a summary by OUT (stock-out) category.

Request Headers

Key	Value
Accept	application/json
Authorization	Bearer <TOKEN>

Request Parameters

Key	Rules
month	optional, integer, between 1 and 12
year	optional, integer

E1a: Response Success

Key	Value
Status Code	200
	{ <pre> "status": "success", "message": "Get summary by out category successful", "data": { "summary": [{ "category": { "id": 9, "name": "Sale", "icon": "🛒", "color": "#55EFC4", "type": "OUT", "created_at": "2025-07-28T02:37:46.000000Z", "updated_at": "2025-07-28T02:37:46.000000Z" }, "quantity": 350 }, ...] } </pre>

E1b: Response Invalid Token

Key	Value
Status Code	401
	{ <pre> "status": "error", "message": "Unauthenticated." </pre>

E2: Get Summary by IN Category

GET /api/reports/summary-by-category/in

For logged-in users to get a summary by IN (stock-in) category.

Request Headers

Key	Value
Accept	application/json

Key	Value
Authorization	Bearer <TOKEN>

Request Parameters

Key	Rules
month	optional, integer, between 1 and 12
year	optional, integer

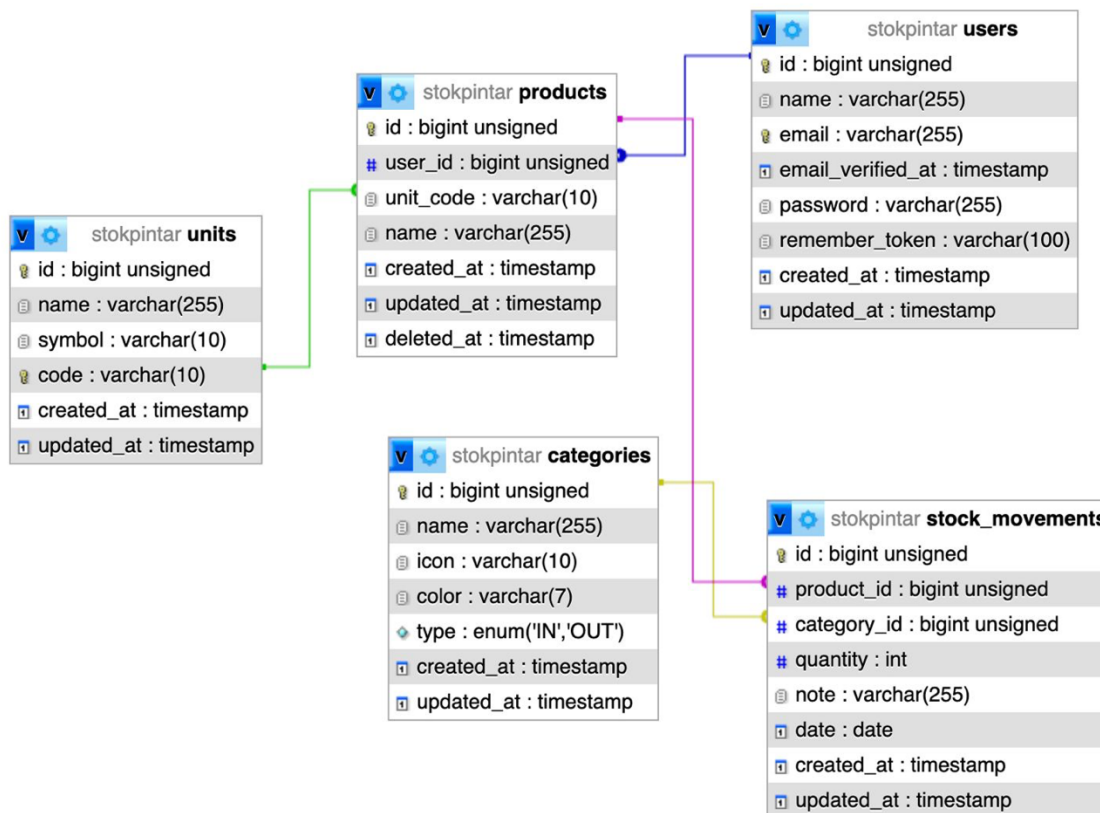
E2a: Response Success

Key	Value
Status Code	200
	<pre>{ "status": "success", "message": "Get summary by in category successful", "data": { "summary": [{ "category": { "id": 1, "name": "Purchase", "icon": "📦", "color": "#00B894", "type": "IN", "created_at": "2025-07-28T02:37:46.000000Z", "updated_at": "2025-07-28T02:37:46.000000Z" }, "quantity": 1200 }, "..."] } }</pre>

E2b: Response Invalid Token

Key	Value
Status Code	401
	<pre>{ "status": "error", "message": "Unauthenticated." }</pre>

ER DIAGRAM



MEDIA FILES

- **Postman Collection & Environment** (postman-collection/) – contains all required API endpoints with examples
- **Database Dump (db-dump.sql)** – MySQL schema and sample data, used as reference
- **Laravel 12 Starter (laravel12-server-phase-1.zip)** – pre-scaffolded Laravel 12 + Sanctum + 5 seeders, vendor/ pre-installed (offline-ready). Uses MySQL. See README.md inside.

INSTRUCTIONS

- You are not allowed to create additional endpoints.
- You can add extra fields to the response body if needed, just make sure all required fields are included and the response structure is correct.